

🌀 M002 · While Loop

Topic: Repeat Until a Wall · **Course:** Maze Madness · **Level:** Advanced · **Time:** about 55 minutes

A while loop checks, runs the body, then checks again. It stops when the check is false.

1 · Trace the Loop 🧠

```
while not agent.detect(BLOCK, FORWARD):  
    agent.move(FORWARD, 1)
```

The wall is 5 steps ahead. How many moves? When does the loop end?

2 · Read the API

Call	What it does
<code>agent.detect(BLOCK, FORWARD)</code>	true if a wall is ahead
<code>agent.move(FORWARD, 1)</code>	step forward once
<code>agent.turn(RIGHT_TURN)</code>	turn right

Why check `not agent.detect(...)` and not `agent.detect(...)` ?

3 · Spot the Bugs 🐛

Bug A — should move until the wall.

```
while agent.detect(BLOCK, FORWARD):  
    agent.move(FORWARD, 1)
```

Fix it:

Bug B — should stop right at the wall, not crash.

```
while not agent.detect(BLOCK, FORWARD):  
    agent.move(FORWARD, 1)  
    agent.move(FORWARD, 1)
```

Fix it. Say why it crashed:

4 · Write a Chat Command

Make `go` move the agent forward until a wall.

Use `player.on_chat` , `while` , `agent.detect` , `agent.move` .

5 · Loop, Turn, Loop

Move to the wall, turn right, then move to the next wall.

6 · Show Your Work 📷 🎥

Open your homework world. Use a while loop to drive the agent down the straight maze to the wall.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.